JACKSON MORRIS

CURRICULUM VITAE

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EMPLOYMENT

Vanderbilt University — Nasvhille, TN
Postdoctoral Scholar — *2027-2029*

Simon Laufer Mathematical Sciences Institute — Berkeley, CA
Postdoctoral Scholar — *Fall 2026*
-*Program*: Motivic Homotopy Theory: Connections and Applications

EDUCATION

University of Washington — Seattle, WA
Ph.D. in Mathematics — *2021-2026*
-*Advisors*: Kyle Ormsby and John Palmieri
-*Thesis title*: Cooperations in motivic homotopy theory
M.S. in Mathematics

University of Kentucky — Lexington, KY
B.S. in Mathematics *magna cum laude* — *2017-2020*

RESEARCH INTERESTS

Motivic homotopy theory, chromatic homotopy theory, hermitian K-theory, algebraic K-theory, splittings, synthetic spectra, nilpotence and periodicity, Adams spectral sequence, periodic self-maps, redshift and blueshift

PUBLICATIONS AND PREPRINTS

10. *Derived Galois and fracture models for cellular motivic spectra over* $\mathbb{Q}_2$, joint with Luca De Paulis, Andrea Figini, Marco Giustetto, and Paul Arne Østvær. In preparation.
9. *Algebraic K-theory of Stiefel varieties*, joint with Sebastian Gant. In preparation.
8. *Motivic K-theory cooperations over the rationals,* joint with Sarah Petersen and Liz Tatum. In preparation.
7. *A* $\mathbb{C}$*-motivic* w_1*-self-map of periodicity 1.* Submitted. Available at https://arxiv.org/abs/2610.22651
6. *Periodic phenomena in stable motivic homotopy theory.* Submitted. Available at https://arxiv.org/abs/2607.18165
5. *Immersions of* C_2*-projective spaces via* $\mathrm{K}\mathbb{R}$*-theory*, joint with Manyi Guo, Alex Waugh, and Albert Jinghui Yang. Submitted. Available at https://arxiv.org/abs/2604.25260.
4. *Spectrum level splittings of truncated motivic Brown–Peterson cooperations algebras*, joint with Sarah Petersen and Elizabeth Tatum. Submitted. Available at https://arxiv.org/abs/2509.19542.
3. *Rings of cooperations for hermitian K-theory over finite fields.* To appear in **Algebraic & Geometric Topology**. Available at https://arxiv.org/abs/2509.02786.
2. *On the ring of cooperations for real hermitian K-theory.* **Annals of K-theory 11** (2026), **4**, 633-708.
1. *Toric Double Determinantal Varieties*, joint with Alexander Blose, Patricia Klein, and Owen McGrath. **Communications in Algebra 49** (2020), **7**, 3085-3093.

TEACHING EXPERIENCE

Instructor of Record

Vanderbilt University
Math 2410 – Methods of linear algebra *Spring 2027*
Math 2610/5610 – Ordinary differential equations *Spring 2027*

University of Washington
Math 480 – Homotopy theory *Spring 2026*
Math 208 – Linear algebra *Summer 2023*
Math 208 – Linear algebra *Winter 2023*

Graduate Assistant

Electronic Computational Homotopy Theory (eCHT)
Research Seminar *2025–2026*
Quadratic Curve Counting *Fall 2024*

Park City Math Institute (PCMI) *Summer 2024*
Experimental Math Lab

University of Washington *2021–2026*
Courses included: Precalculus, Differential calculus, Multivariable calculus, Linear algebra, Introduction to proofs, Topology

Advising/Mentoring

Directed Reading Mentor UW
Topics included: Homotopy theory, Representation theory, Geometric group theory, Homological algebra *2022–2025*

Advising/Menthorship

Jason Yeung (University of Toronto) SLMath
Program associate mentor during thematic program *Fall 2026*
Spencer Catron (University of Washington) UW
Graduate student mentor during first year *2023-2024*

Math Olympiad Judge

UW Math Hour *2023–2024*
Judge for elementary school math competition

HONORS AND AWARDS

Lee and Cozette McFarlan Travel Fellowship *Summer 2025*
University of Washington *$750*
For travel to European Talbot in Kolding, DK.

Research Assistantship *Summer 2025*
University of Washington
For good research progress.

JC Eaves Scholarship *2020*
University of Kentucky *$2000*
For continued excellence in mathematics as a Kentuckian.

Carolyn Bunyan Award *2020*
University of Kentucky *$2000*
For mathematics majors interested in continuing education in mathematics.

SERVICE AND PROFESSIONAL ACTIVITIES

Refereeing and Reviewing

Algebraic and Geometric Topology *2025-Present*
ZBMath *2026-Present*

Seminar Organization

Learning Seminar SLMath
Motivic stable stems and Galois approximations of cellular motivic categories *Fall 2026*

DUBTOP Seminar UW
Topics included: Chromatic homotopy theory, stacks, Goodwillie calculus *2024-2025*

Back-To-School seminar UW
Student lectures on fun math learned over the summer *Fall 2024*

Reading seminars UW
Topics included: Algebraic K-theory, cohomology operations, motivic homotopy theory, vector bundles and topological K-theory, categorical homotopy theory, model categories *2023-2024*

SELECTED PRESENTATIONS & SEMINAR TALKS

Conference and Colloqium Talks

AMS Sectional: Computational Homotopy Theory Tempe, AZ
$\mathbb{C}$*-motivic periodic self-maps* *Novermber 2026*
Motivic Homtopy Theory: Connections and Applications SLMath, Berkeley, CA
Motivic K-theory cooperations over the rationals *October 2026*
Young Topologists Meeting Copenhagen, DK
Higher Witt K-theories *July 2026*
Cascade Topology Seminar Portland, OR
Higher Witt K-theories *May 2026*
Joint Mathematics Meetings 2026 Washington, D.C.
Splittings and cooperations in motivic homotopy theory *January 2026*
Graduate Student Geometry & Topology Seminar Bloomington, IN
On the ring of cooperations for real hermitian K-theory *April 2025*
Reed Mathematics Colloquium Portland, OR
Finding Fixed Points with Invariants *February 2025*
Joint Mathematics Meetings 2025 Seattle, WA
On the ring of cooperations for real hermitian K-theory *January 2025*

Seminar Talks

New Mexico State University Topology Seminar Las Cruces, NM
$\mathbb{C}$*-motivic periodic self-maps* *September 2026*
eCHT Research Seminar Online
$\mathbb{C}$*-motivic periodic self-maps* *September 2026*
University of Washington Topology Seminar Seattle, WA
Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras *February 2026*
University of Illinois Urbana-Champaign Topology Seminar Champaign, IL
Splittings and cooperations in motivic homotopy theory *January 2026*
University of Virginia Topology Seminar Charlottesville, VA
Splittings and cooperations in motivic homotopy theory *November 2025*
University of Kentucky Topology Seminar Lexington, KY
Splittings and cooperations in motivic homotopy theory *November 2025*
Duke University Geometry and Topology Seminar Durham, NC
Splittings and cooperations in motivic homotopy theory *October 2025*
University of Colorado Boulder Homotopy Theory Seminar Boulder, CO
Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras *September 2025*
University of Michigan Geometry Seminar Ann Arbor, MI
On the ring of cooperations for real hermitian K-theory *September 2025*

University of Notre Dame Topology Seminar — South Bend, IN
Splittings and cooperations in motivic homotopy theory — *September 2025*
University of Washington Topology Seminar — Seattle, WA
On the ring of cooperations for real hermitian K-theory — *June 2025*

Contributed Talks

Reading Seminars
An upper bound on the degree of symmetry of an exotic sphere — *March 2026*
Higher Adams differentials and hidden extensions — *November 2025*
Motivic twisted K-theory — *April 2024*
Higher real K-theories, redshift, and blueshift — *May 2026*
Filtered objects, spectral sequences, and t-structures — *January 2026*
(A synthetic approach to detecting) v_1-periodic families — *November 2025*
Algebraic and Milnor K-theory — *October 2025*
Galois descent and the Picard group of K-theory — *May 2025*
The Adams spectral sequence and Hopf algebroids — *February 2025*
Symplectic orientations — *August 2024*

Workshops
$\pi_* THH(\mathbb{Z}_p^{fil})/p$ *and* $\pi_* THH(\ell_p^{fil})/(p, v_1)$ (Talbot, Nacogdoches TX) — *August 2024*
Hochschild Homology (European Autumn School in Topology, Utrecht, NL) — *September 2024*